

CHAPTER 1

INTRODUCTION

1.1. Background and Motivation

Carbon dioxide (CO₂) is one of the major greenhouse gases responsible for global warming and climate change. Rapid industrialization, especially in sectors like cement production, power generation, and manufacturing, has led to a significant increase in CO₂ emissions. These emissions not only affect the environment but also contribute to rising global temperatures and air pollution. Traditional Carbon Capture and Storage (CCS) systems are widely used in large-scale industries to reduce CO₂ emissions. However, these systems are expensive, require large infrastructure, and are not feasible for small and medium-scale industries. At the same time, the construction industry relies heavily on conventional concrete curing methods, which do not utilize CO₂ effectively. Recent advancements have shown that CO₂ can be used in the carbonation curing process of concrete, improving its strength and durability while reducing environmental impact. Integrating this concept with IoT technology enables real-time monitoring and efficient management of CO₂ levels, making the system smarter and more efficient.

1.2. Need for the Project

There is an urgent need to develop sustainable and cost-effective solutions to reduce carbon emissions from industries. Existing carbon capture technologies are not accessible to small-scale industries due to their high cost and complexity. Additionally, conventional concrete curing processes do not make use of CO₂, which is a missed opportunity for carbon utilization. There is also a lack of systems that can monitor CO₂ levels in real time during industrial processes. Therefore, this project is needed to:

- Provide a low-cost and compact CO₂ capture system.
- Utilize captured CO₂ for concrete curing.
- Enable real-time monitoring using IoT technology.
- Support environmentally friendly and sustainable construction practices.

1.3. Problem Statement

Industrial processes release large amounts of carbon dioxide into the atmosphere, contributing to environmental pollution and global warming. Existing carbon capture systems are costly, complex, and designed mainly for large-scale industries. There is no integrated system that combines CO₂ capture, utilization in concrete curing, and real-time monitoring in a compact and affordable manner. Small and medium-scale industries lack access to such efficient and sustainable solutions.

1.4. Objective of the Project

- To design and develop a controlled system for CO₂ capture, transfer, and carbonation curing of concrete.
- To improve the strength and durability of concrete through effective CO₂ curing techniques.
- To implement a low-cost, compact prototype suitable for small-scale industrial applications.
- To integrate IoT-based real-time monitoring of CO₂ concentration using sensors and an ESP32 platform for sustainable construction practices.

1.5. Scope of the Project

The scope of this project includes the design and development of a prototype system that integrates CO₂ capture, carbonation curing, and IoT-based monitoring. The system is mainly intended for small and medium-scale industries and precast concrete manufacturing units. It focuses on capturing CO₂ from a

controlled source and utilizing it effectively in the curing process of concrete blocks. The project also includes real-time monitoring of CO₂ levels using sensors and an ESP32-based IoT platform. However, the current work is limited to a prototype model and does not cover large-scale industrial implementation. Future enhancements can expand this system for industrial applications, improve automation, and integrate advanced data analytics for better performance.

1.6. Innovation

This project presents a smart and practical way to reduce carbon dioxide (CO₂) pollution by turning it into something useful. Instead of releasing CO₂ into the air, the system captures it and uses it during the curing process of concrete blocks. This not only helps in reducing pollution but also improves the strength and durability of the concrete. Unlike existing carbon capture systems that are costly and used only in large industries, this project focuses on a low-cost and compact design that can be used in small and medium-scale applications. Another unique feature is the use of an ESP32-based IoT system, which allows real-time monitoring of CO₂ levels before and after curing. By combining environmental protection with construction benefits, this project offers a simple, effective, and sustainable solution for modern industrial and construction challenges.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Existing Systems

Carbon Capture and Storage (CCS) and Carbon Capture, Utilization, and Storage (CCUS) systems are widely used to reduce carbon dioxide (CO₂) emissions from industrial sources. These systems typically capture CO₂ from exhaust gases, compress it, and store it in underground geological formations such as saline aquifers or depleted oil and gas reservoirs. In addition, some systems focus on Carbon Capture and Utilization (CCU), where CO₂ is used in industrial processes like enhanced oil recovery, chemical production, and concrete curing. Recently, IoT-based monitoring systems have also been developed to track emissions using sensors and cloud platforms. However, most of these systems are designed for large-scale industries and require complex infrastructure, making them unsuitable for small and medium-scale industries.

2.2 Roy, P., Mohanty, A. K., & Misra, M. (2023). Prospects of carbon capture, utilization and storage for mitigating climate change. Environmental Science: Advances, 2(3), 409-423.

This paper presents a comprehensive study on Carbon Capture, Utilization, and Storage (CCUS) technologies aimed at reducing global carbon dioxide (CO₂) emissions. It highlights the urgent need to achieve net-zero emissions by 2050 to control global temperature rise. The study discusses various CO₂ capture methods such as pre-combustion, post-combustion, and oxy-fuel combustion. It also explores storage techniques including geological storage in saline aquifers, depleted oil and gas reservoirs, and mineral carbonation. Additionally, the paper emphasizes the importance of CO₂ utilization in producing fuels, chemicals, and construction materials.

The key contributions of the paper include a detailed explanation of CCUS technologies, a comparison of different CO₂ capture methods, and insights into global carbon reduction strategies. It strongly focuses on sustainability and environmental protection. However, the study mainly targets large-scale industrial applications and requires high investment and complex infrastructure. It does not address compact or low-cost solutions and lacks integration of real-time monitoring systems using IoT technologies.

Overall, the paper provides a strong theoretical foundation for understanding CO₂ capture and utilization. At the same time, it reveals a gap in developing smaller, cost-effective, and smart systems. This limitation motivates the proposed project, which aims to design a compact, affordable, and IoT-enabled solution for CO₂ capture, utilization, and monitoring.

2.3 Herraiz, L., Moser, P., Wiechers, G., Stoffregen, T., Putta, K. R., Morgan, J. C., ... & Garcia, S. (2025). Techno-economic evaluation of emission control configurations for AMP/PZ-based post-combustion CO₂ capture. Chemical Engineering Journal, 171446.

This paper focuses on the performance and economic analysis of amine-based CO₂ capture systems using AMP (2-Amino-2-methyl-1-propanol) and PZ (Piperazine) solvents. These systems are widely used in post-combustion capture processes, especially in thermal power plants. The study examines different configurations such as absorber and stripper units, and evaluates key factors including energy consumption, solvent degradation, and operational costs. It also highlights environmental concerns like ammonia emissions and aerosol formation during the capture process.

The key contributions of the paper include a detailed techno-economic analysis of CO₂ capture systems and an evaluation of solvent-based capture efficiency. It identifies important operational challenges and environmental risks while emphasizing emission control and regulatory compliance. However, the system

is designed mainly for large-scale power plants and involves high capital investment and maintenance costs. The overall system design is complex and difficult to implement on a smaller scale, and it does not include CO₂ utilization or IoT-based monitoring.

Overall, the paper demonstrates the effectiveness of advanced CO₂ capture techniques but also highlights their complexity and high cost. These limitations indicate the need for a simpler and more affordable solution. This supports the motivation behind the proposed project, which focuses on developing a low-cost, compact, and IoT-enabled system suitable for small and medium-scale industries.

2.4. Liu, J., & Zhang, Z. (2023). Integrated energy carbon emission monitoring and digital management system for smart cities. *Frontiers in Energy Research*, 11, 1221345.

This paper presents an IoT-based system for monitoring carbon emissions in smart cities using wireless sensor networks. It collects real-time data on CO₂ emissions from various sources such as buildings, transportation, and industrial areas. The system integrates advanced technologies like cloud computing and data analytics to improve the efficiency and accuracy of carbon emission tracking compared to traditional estimation methods. It also supports better environmental management and contributes to smart city development.

The key contributions include real-time emission monitoring, integration of sensor networks with cloud platforms, and improved data accuracy. However, the system focuses only on monitoring and does not include CO₂ capture or utilization. It lacks emission control mechanisms and is not specifically designed for industrial applications. This highlights the need for a more practical solution, which is addressed in the proposed project by combining IoT-based monitoring with CO₂ capture and utilization for effective emission reduction.

2.5. Fortunato, L. R., Parsekian, G. A., & Neves, A. (2022). Feasibility analysis for implementing CO₂ curing in a concrete block industry in the São Paulo Region. Revista IBRACON de Estruturas e Materiais, 15(6), e15607.

This paper presents a feasibility analysis for implementing CO₂-based carbonation curing in a concrete block industry, particularly in the São Paulo region. It explains how CO₂ can be used during the curing process of concrete blocks, where it reacts with cement compounds to form calcium carbonate (CaCO₃), leading to permanent carbon storage and improved mechanical strength. The study also discusses the technical requirements such as curing chambers, CO₂ supply, temperature, humidity, and pressure control. It highlights that this method is a sustainable innovation that reduces curing time, improves productivity, and lowers environmental impact by utilizing industrial CO₂ emissions.

The paper concludes that implementing CO₂ curing is technically feasible with only minor modifications to existing production systems and curing chambers. Although there may be a slight increase in production cost (around 4% to 14%), the environmental benefits are significant, with the potential to sequester large amounts of CO₂ annually. However, the study mainly focuses on industrial-scale implementation and does not address low-cost or IoT-based monitoring solutions. This creates an opportunity for developing compact, smart systems like the proposed project, which combines CO₂ capture, utilization, and real-time monitoring.

2.6 Liu, Z. (2024). Effect of Carbon Dioxide Curing on the Carbonation Durability of Concrete. Frontiers in Sustainable Development.

This paper studies the effect of carbon dioxide (CO₂) curing on the durability and performance of concrete. It explains how CO₂ curing works by reacting with cement compounds such as calcium hydroxide to form calcium carbonate (CaCO₃), which fills the pores of concrete and makes it denser and stronger. The study highlights that this method helps in carbon sequestration and improves early strength, making it a promising technique for reducing carbon emissions in the construction industry. It also discusses factors like CO₂ concentration, temperature, humidity, and curing time that influence the efficiency of the carbonation process.

However, the paper also points out some important limitations. CO₂ curing can reduce the pH of concrete, which may lead to corrosion of steel reinforcement and affect long-term durability. It also shows that although CO₂ curing increases carbon storage (around 8%–23%), it may reduce the service life of concrete structures due to faster carbonation. Additionally, energy consumption during CO₂ capture and processing may reduce overall environmental benefits. Therefore, while CO₂ curing is effective for carbon capture and strength improvement, further research is needed to balance durability and sustainability.

2.7 Limitations of Existing Systems

From the literature review, the following limitations are identified:

- Most systems are designed for large-scale industries
- High installation and operational costs
- Complex infrastructure and maintenance
- Lack of integration between capture, utilization, and monitoring
- Limited accessibility for small and medium-scale industries (MSMEs)

- Absence of compact and portable solutions
- IoT-based systems focus only on monitoring, not emission control

2.8 Research Gap Identified

- Most of the existing systems are designed for large industries and are costly, so they are not suitable for small-scale or local applications.
- Many studies focus only on either CO₂ capture or monitoring, but very few combine capture, usage, and monitoring together in one system.
- Even though CO₂ curing improves concrete strength, there is a lack of simple and affordable systems that can be easily implemented.
- Real-time monitoring using IoT technology is missing in most of the existing works, making it difficult to track CO₂ levels effectively.
- Most research is theoretical or large-scale, with very limited work on small working models or prototypes.
- Some studies mention problems like durability and environmental impact, but they do not provide practical solutions to balance all factors.

CHAPTER 3

PROPOSED METHODOLOGY

3.1. Overview

The proposed system is an IoT-enabled CO₂ capture and carbonation curing system designed to reduce carbon emissions and utilize CO₂ effectively in the construction process. The system captures carbon dioxide from a controlled source and directs it into a curing chamber containing precast concrete blocks. During the curing process, CO₂ reacts with the concrete and enhances its strength and durability through carbonation. An IoT-based monitoring system using ESP32 is integrated to track CO₂ concentration levels in real time. The system is compact, cost-effective, and suitable for small and medium-scale industries, addressing the limitations of existing large-scale carbon capture systems.

3.2. System Architecture

The system architecture is represented in Fig 3.1 which is designed in such a way that CO₂ is captured, controlled, and utilized efficiently while continuously monitoring the system performance through OLED Display.

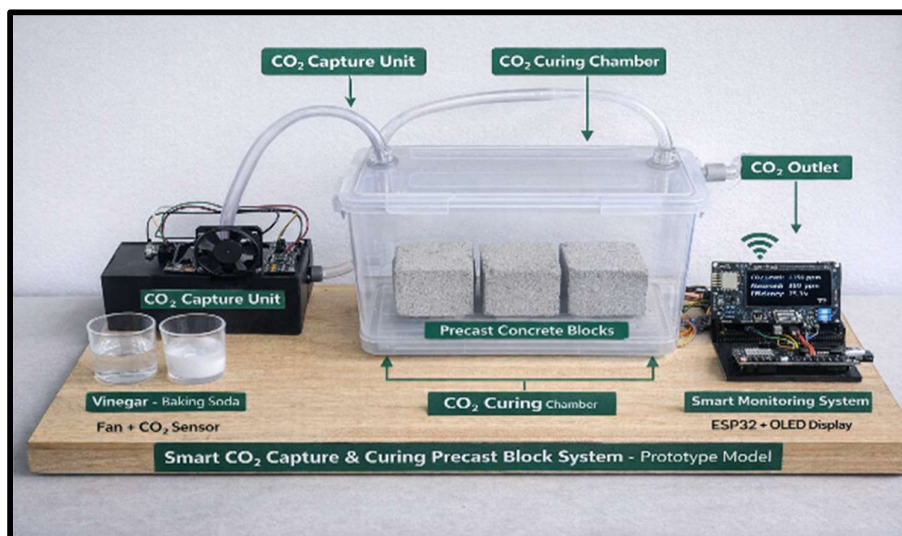


Figure 3.1 System architecture

3.3. Block Diagram

The Fig 3.3 represents the Block diagram of the project.

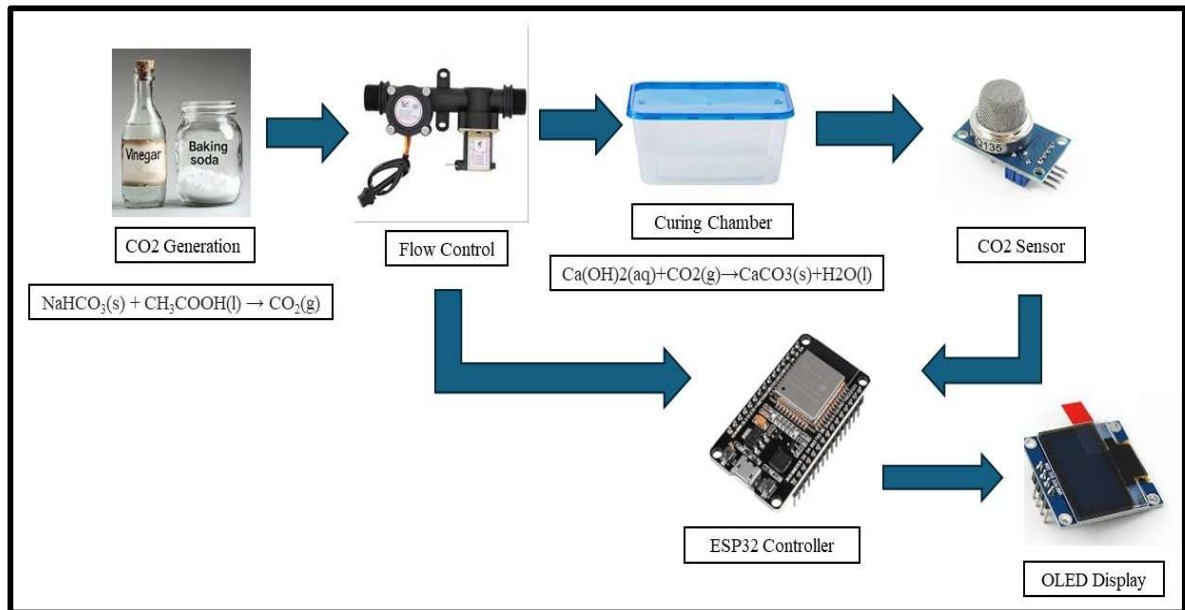


Figure 3.2: Block Diagram

3.3.1. Block Diagram Description:

The block diagram of the system includes the following stages:

1. CO₂ Source: Generates or provides carbon dioxide.
2. Capture Unit: Collects CO₂ gas.
3. Flow Control Unit: Regulates gas flow using valves.
4. Curing Chamber: Contains concrete blocks for carbonation.
5. Gas Sensor: Measures CO₂ concentration.
6. ESP32 Module: Processes sensor data.
7. OLED Display: Displays real-time data.

Each block works together to ensure proper capture, transfer, utilization, and monitoring of CO₂ in the system.

3.4 Flow Chart

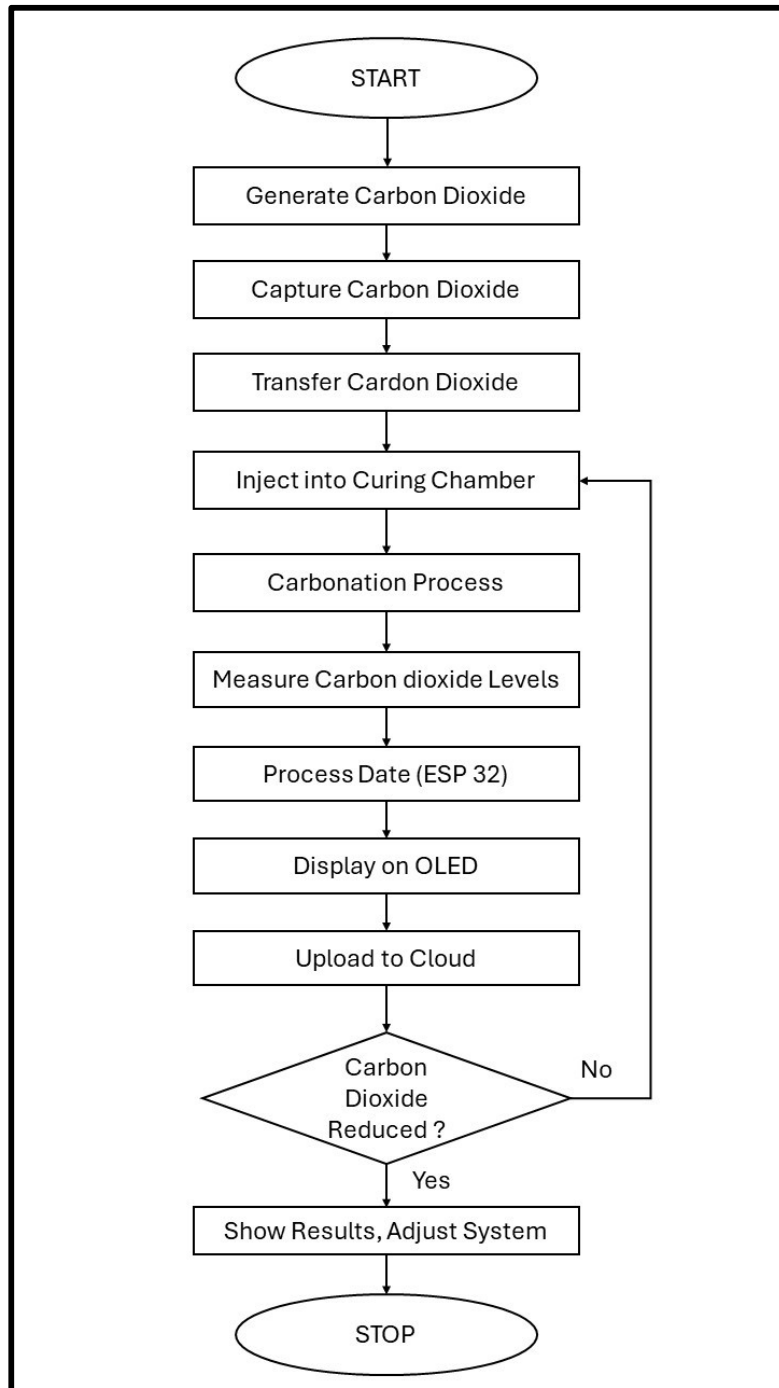


Figure 3.3 Flow chart

The overall process flow of the system can be summarized in Fig 3.3

- CO₂ Capture → Gas Transfer → Flow Control → CO₂ Curing → Gas Sensing → OLED Display

3.5 Working Principle

The working of the system is based on the concept of carbonation curing and IoT monitoring, and it operates as follows:

1. CO₂ is generated from a chemical reaction or external source.
2. The gas is captured and directed into a pipeline.
3. Flow control valves regulate the movement of CO₂.
4. CO₂ enters the curing chamber containing concrete blocks.
5. Carbonation reaction occurs between CO₂ and concrete.
6. This reaction improves the strength and durability of concrete.
7. A CO₂ sensor continuously measures gas concentration.
8. ESP32 collects sensor data and monitored through OLED Display.
9. Real-time monitoring and analysis are performed.

3.5 Key Features

- Integrated CO₂ capture and utilization system.
- Real-time monitoring using IoT.
- Low-cost and compact design.
- Suitable for small-scale industries.
- Enhances concrete strength through carbonation.
- Environmentally friendly and sustainable.

3.7. Testing and Validation

After constructing the curing chamber and OLED display system, the entire setup is tested to ensure proper performance and reliability. The MQ135 sensor is calibrated to measure CO₂ levels before and after the curing process, confirming effective carbon capture through a reduction in CO₂ concentration. The gas transfer components, including the fan, valve, and pipes, are tested to ensure smooth and leak-free CO₂ flow into the chamber. A compressive strength test is

conducted to evaluate the strength of the concrete blocks, showing improved durability compared to conventional curing. Additionally, the ESP32-based OLED display is validated to ensure accurate real-time monitoring and proper data visualization, confirming the overall effectiveness of the system.

3.8. Maintenance and Monitoring

Regular maintenance ensures efficient system performance and long-term reliability. The MQ135 sensor is periodically calibrated for accurate CO₂ measurement, and components like the fan, valve, and pipelines are checked to prevent leakage. The curing chamber is inspected to maintain proper conditions. The ESP32-based monitoring system continuously monitors CO₂ levels and displays real-time data on the OLED, helping detect issues quickly and ensure smooth operation.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1. Hardware Requirements

The hardware components used in this project are selected based on cost-effectiveness, availability, and ease of integration. The following are the major hardware requirements:

4.1.1. ESP32 Microcontroller

The ESP32 is a low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth, widely used in IoT applications. It is developed by Espressif Systems and is ideal for real-time data monitoring and control systems. In this project, the ESP32 represented in Fig 4.1 is used to collect data from the MQ135 CO₂ sensor and processes it to display on the OLED screen and IoT platform. It enables wireless data transmission, allowing users to monitor CO₂ levels remotely. With its high processing capability, multiple GPIO pins, and energy efficiency, the ESP32 plays a key role in ensuring accurate monitoring and smart control of the system.

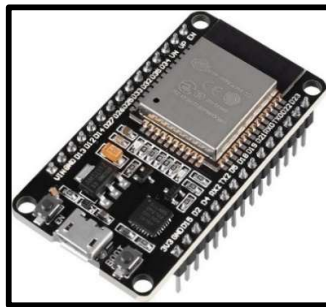


Figure 4.1 ESP32 Microcontroller

4.1.2. CO₂ Gas Sensor (MQ135)

The MQ135 gas sensor represented in Fig 4.2 is used to detect air quality and measure gases such as CO₂, NH₃, NO_x, and smoke. In this project, it is mainly

used to monitor the concentration of CO₂ inside the curing chamber. The sensor works based on changes in resistance when exposed to different gas levels, providing analog output that can be read by the ESP32. It is cost-effective, sensitive, and suitable for continuous monitoring. Proper calibration is required to ensure accurate CO₂ measurement and reliable performance in the system.



Figure 4.2 CO₂ Gas Sensor

4.1.3. Curing Chamber

The curing chamber is a sealed enclosure where concrete blocks are exposed to controlled CO₂ for the carbonation curing process. It allows CO₂ to react with calcium hydroxide in concrete to form calcium carbonate, improving strength and durability. The chamber is designed to prevent gas leakage and maintain proper curing conditions such as uniform CO₂ distribution. It plays a key role in enhancing concrete properties while enabling effective carbon sequestration.

4.1.4. CO₂ Capture Unit

The CO₂ capture unit is responsible for collecting carbon dioxide from a controlled source, such as a chemical reaction setup. It ensures that CO₂ is safely captured and directed into the curing chamber without loss. The unit includes components like a reaction container, gas outlet pipe, and flow control mechanism to regulate CO₂ supply. It plays a crucial role in providing a continuous and controlled flow of CO₂ for effective carbonation curing.

4.1.5. Gas Transfer Pipes

The gas pipelines are used to safely transport CO₂ represented in Fig 4.3 from the capture unit to the curing chamber. They ensure a smooth and controlled flow of gas without leakage. The pipelines are made of durable, airtight materials and are properly sealed to maintain efficiency. They play an important role in connecting different components of the system and ensuring effective CO₂ transfer.



Figure 4.3 Gas Transfer Pipes

4.1.6. Flow Control Valves

The flow control valve represented in Fig 4.4 regulates the amount of CO₂ entering the curing chamber. It allows precise control of gas flow to maintain optimal conditions for carbonation curing. By adjusting the valve, the CO₂ supply can be increased or decreased as required, preventing overpressure and ensuring safe operation. It plays a key role in maintaining system efficiency and controlled gas distribution.



Figure 4.4 Flow Sensor

4.1.7.OLED Display

The OLED display represented in Fig 4.5 is used to show real-time data such as CO₂ concentration and system status. It provides a clear and efficient visual output with low power consumption. In this project, it is connected to the ESP32 to display sensor readings instantly, helping users monitor the system easily.

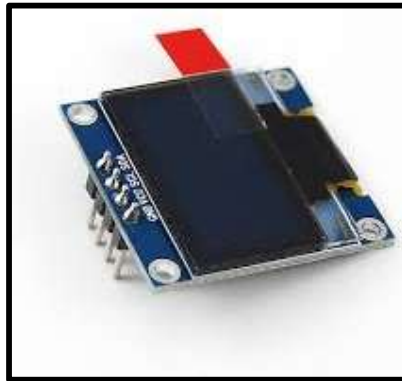


Figure 4.5 OLED Display

4.1.8. Power Supply

Power Supply provides the required electrical energy to operate all system components including ESP32, sensors, fan, and valve. It ensures stable and continuous operation by delivering regulated voltage to the entire system.

4.1.9. DC Fan

The 12V DC fan is used to facilitate the smooth movement of CO₂ gas from the capture unit to the curing chamber. It ensures proper airflow and uniform distribution of gas inside the chamber. The fan helps in maintaining continuous gas circulation, improving the efficiency of the carbonation process.

4.2. Software Requirements

The software components are used for programming, data processing, and IoT monitoring.

4.2.1. Arduino IDE

The Arduino IDE (Integrated Development Environment) is an open-source software platform used for writing, compiling, and uploading code to microcontrollers like the ESP32. It provides an easy-to-use interface with a built-in code editor, compiler, and serial monitor for debugging. In this project, the Arduino IDE is used to program the ESP32 to read CO₂ levels from the MQ135 sensor, control components like the fan and valve, and display data on the OLED screen. The availability of various libraries simplifies sensor integration, display handling, and make development faster and more efficient.

4.2.2 Program

```
#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT,&Wire, -1);

#define MQ135_IN 34

#define MQ135_OUT 35

#define FAN_RELAY 26

#define VALVE_RELAY 27

#define FLOW_SENSOR 32

volatile int flowPulse = 0;
```

```

float flowRate = 0;

void IRAM_ATTR flowInterrupt() {

    flowPulse++;

}

void setup() {

    Serial.begin(115200);

    pinMode(FAN_RELAY, OUTPUT);

    pinMode(VALVE_RELAY, OUTPUT);

    pinMode(FLOW_SENSOR, INPUT);

    attachInterrupt(digitalPinToInterrupt(FLOW_SENSOR),flowInterrupt,RISING);

    digitalWrite(FAN_RELAY, LOW);

    digitalWrite(VALVE_RELAY, LOW);

    if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {

        Serial.println("OLED not found");

        while (true);

    }

    display.clearDisplay();

    display.setTextColor(WHITE);

}

void loop() {

    int sensorIn = analogRead(MQ135_IN);

    int sensorOut = analogRead(MQ135_OUT);

    float inputCO2 = map(sensorIn, 0, 4095, 400, 5000);

    float outputCO2 = map(sensorOut, 0, 4095, 400, 5000);

```

```

digitalWrite(FAN_RELAY, HIGH);

digitalWrite(VALVE_RELAY, HIGH);

flowRate = flowPulse * 0.1;

flowPulse = 0;

Serial.print("Input CO2: ");

Serial.print(inputCO2);

Serial.println(" ppm");

Serial.print("Output CO2: ");

Serial.print(outputCO2);

Serial.println(" ppm");

Serial.print("Flow Rate: ");

Serial.print(flowRate);

Serial.println(" L/min");

Serial.println("-----");

display.clearDisplay();

display.setTextSize(1);

display.setCursor(20, 0);

display.println("CO2 MONITOR");

display.setCursor(0, 15);

display.print("In : ");

display.print(inputCO2, 0);

display.println(" ppm");

display.setCursor(0, 30);

display.print("Out: ");

```

```
display.print(outputCO2, 0);  
display.println(" ppm");  
display.setCursor(0, 45);  
display.print("Flow:");  
display.print(flowRate);  
display.println(" L/m");  
display.setCursor(0, 55);  
if (outputCO2 < inputCO2) {  
    display.print("CO2 Reduced");  
} else {  
    display.print("No Change");  
}  
display.display();  
delay(2000);  
}
```

CHAPTER 5

RESULTS AND DISCUSSION

5.1. Prototype

Fig 5.1 represents the Prototype of our project

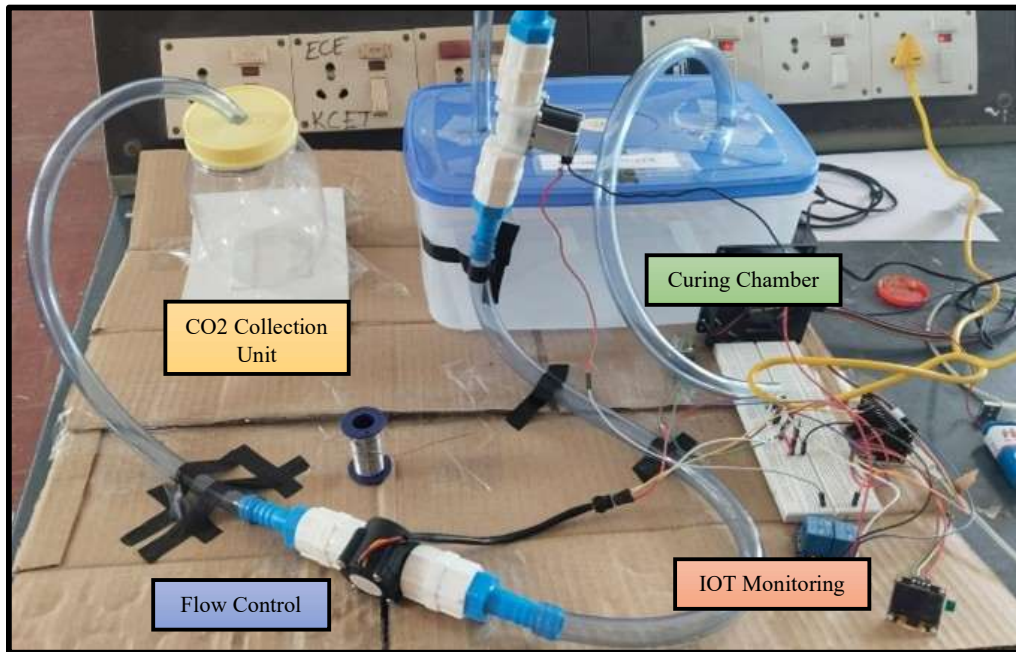


Figure 5.1 Prototype

5.2. Observation

During the experiment, the following observations were made:

- CO₂ gas was successfully generated and captured
- Gas flow was controlled effectively using valves
- CO₂ concentration increased inside the curing chamber
- Sensor readings were stable and responsive
- IoT platform displayed real-time data accurately

5.3 Result

The results is represented in Fig 5.1 which indicate that the system is capable of capturing and utilizing CO₂ effectively. The carbonation curing process showed positive impact on concrete properties. The IoT system provided continuous monitoring, making it easier to track and analyse the data. The system proved to be reliable and efficient for small-scale applications.

CO₂ Level: 900 ppm (before curing) → 400 ppm (after curing)

Flow Rate : 1.5 L/m

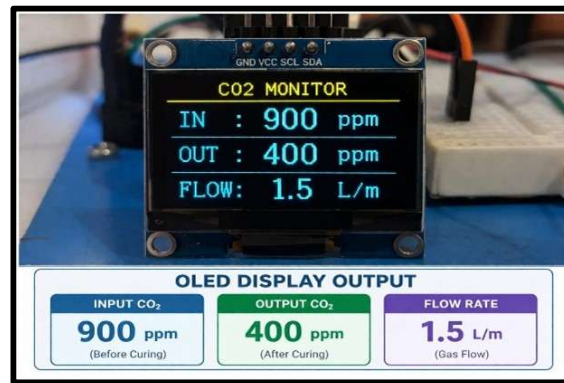


Figure 5.2 Result

5.3.1 Performance Analysis

The performance of the system is evaluated based on:

- Accuracy: Sensor readings were reasonably accurate.
- Efficiency: Effective utilization of CO₂.
- Reliability: Stable operation without failures.
- Response Time: Quick response to gas concentration changes.
- IoT Performance: Smooth data transmission and visualization.

5.3.2. Internal Testing

Internal testing was conducted to evaluate the ability of concrete blocks to absorb CO₂ during the carbonation curing process. The MQ135 sensor measured CO₂ concentration before and after curing inside the sealed chamber, showing a noticeable reduction in CO₂ levels. This confirms that the concrete blocks effectively absorb CO₂ and convert it into calcium carbonate. The controlled environment ensured proper gas exposure, and the results indicate approximately 5–6% CO₂ absorption by cement mass. This validates the efficiency of the system in carbon sequestration within the concrete.

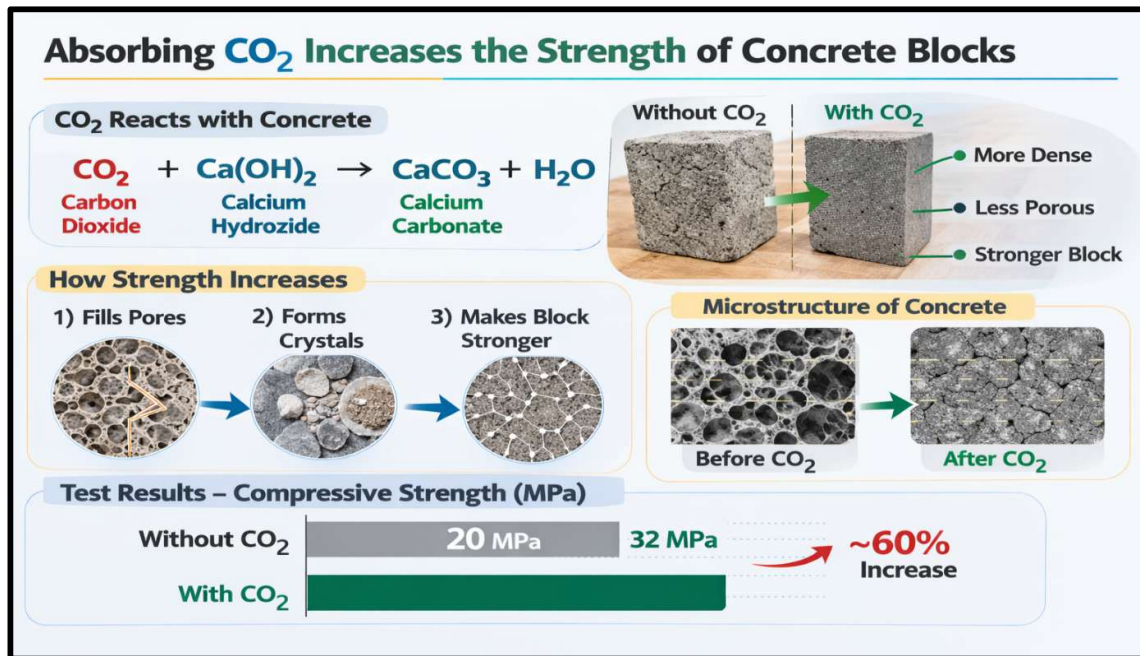


Figure 5.3 Internal Testing

5.4. Discussion

The results indicate that the proposed system effectively integrates CO₂ capture, utilization, and monitoring into a single platform. The reduction in CO₂ concentration confirms successful carbonation curing, while the increase in compressive strength demonstrates improved concrete performance. The use of

fly ash contributes to better durability and reduced cement usage, enhancing sustainability. The IoT-based monitoring system ensures accurate real-time tracking, making the system more reliable and user-friendly. Overall, the approach proves to be a practical, low-cost, and scalable solution for sustainable construction and carbon management.

5.4.1 Strength Analysis

The strength of the concrete blocks is evaluated using a compressive strength test to compare conventional curing with CO₂ carbonation curing. The results show a significant improvement in strength for CO₂-cured blocks, increasing from 21.2 MPa to 29.6 MPa. This enhancement is due to the formation of calcium carbonate within the concrete, which reduces porosity and improves density. The inclusion of fly ash further contributes to strength development through secondary reactions. Overall, the analysis confirms that carbonation curing improves both the strength and durability of concrete blocks.

Strength: 25 MPa → 35 MPa (CO₂ cured concrete)

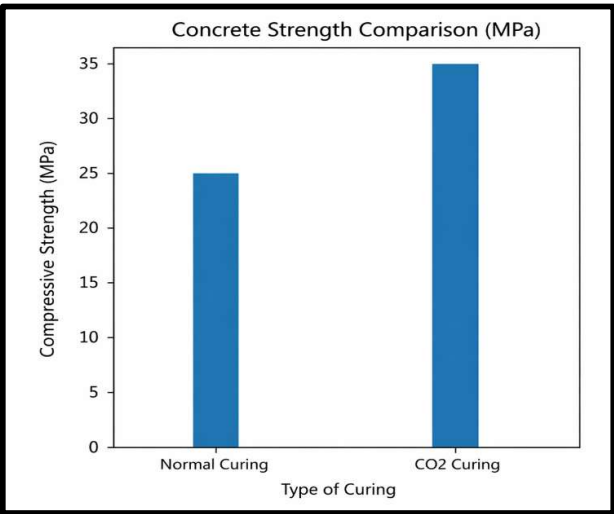


Figure 5.4 Strength Analysis

5.4.2 CO₂ Concentration Analysis

The CO₂ concentration inside the curing chamber is analyzed using the MQ135 gas sensor to evaluate the effectiveness of carbon capture. Measurements are taken before and after the carbonation curing process, showing a clear reduction in CO₂ levels within the chamber. This decrease confirms that CO₂ is successfully absorbed by the concrete and converted into calcium carbonate. The real-time data obtained through the ESP32 and IoT platform ensures accurate monitoring and analysis. Overall, the results validate the efficiency of the system in reducing CO₂ concentration and achieving effective carbon sequestration.

CO₂ Level: 900 ppm → 400 ppm (after curing)

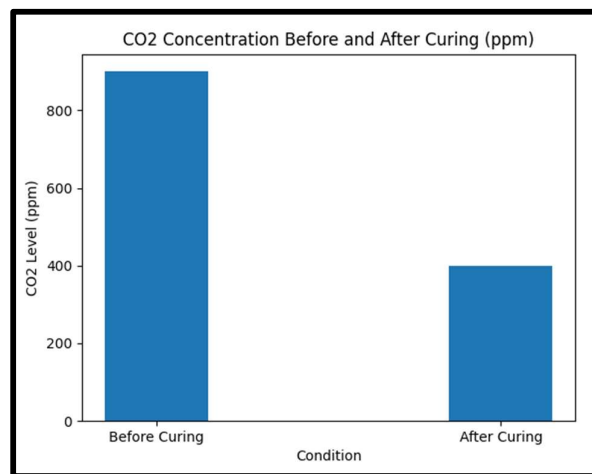


Figure 5.4 CO₂ Concentration Analysis

5.5. Limitation

Although the system performs effectively, it has certain limitations:

- Designed only as a prototype model.
- Limited accuracy due to low-cost sensors.
- Not tested in real industrial conditions.
- Depends on stable Wi-Fi connectivity.
- Limited automation in current implementation.

CHAPTER 6

CONCLUSION

The project titled “IoT-enabled CO₂ Capture and Carbonation Curing System for Sustainable Precast Concrete Blocks” successfully demonstrates an innovative approach to reduce carbon emissions while improving construction materials. The system effectively integrates CO₂ generation, capture, and controlled utilization in concrete curing, proving that captured carbon dioxide can be transformed into a valuable resource. The carbonation process enhances the strength and durability of concrete blocks through the formation of calcium carbonate, thereby achieving both environmental and structural benefits.

Furthermore, the incorporation of IoT technology using the ESP32 microcontroller and CO₂ sensors enables real-time monitoring and efficient data management. The developed prototype is cost-effective, compact, and suitable for small and medium-scale industrial applications. Overall, the project supports sustainable development by promoting eco-friendly construction practices and reducing greenhouse gas emissions, making it a promising solution for modern environmental challenges.

The system can be further improved by integrating advanced high-precision sensors and automation for better control of CO₂ flow and curing conditions. Scaling up the prototype for large-scale industrial applications, along with the use of renewable CO₂ sources, can enhance its real-world impact. Additionally, incorporating AI-based data analysis and cloud integration can improve monitoring efficiency, predictive maintenance, and overall system performance for smarter and more sustainable operations.

CHAPTER 7

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CHAPTER 8

OUTCOMES OF THE PROJECT

8.1 PO Mapping

The following outcomes are attained through the project titled “IoT-Enabled CO₂ Capture and Carbonation Curing System for Sustainable Precast Concrete Blocks”,

S.NO	PO/ PSO	Engineering Graduate will be able to :	Program Outcome Relevance to the Project
1	PO1	Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.	Applied engineering knowledge in IoT, sensors, and concrete technology for CO ₂ capture system.
2	PO2	Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).	Analysed CO ₂ emission problem and identified sustainable solution using carbonation curing.
3	PO3	Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)	Designed an integrated system combining CO ₂ capture, curing chamber, and IoT monitoring.
4	PO4	Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).	Conducted experiments, analysed CO ₂ levels, and validated results with strength testing.
5	PO5	Engineering tool usage: Create, select and apply appropriate	

		techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)	Used modern tools like ESP32, Arduino IDE, sensors, and IoT platforms.
6	PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).	Addressed environmental impact by reducing CO ₂ emissions and promoting sustainability
7	PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)	Followed ethical practices in environmental protection and resource utilization
8	PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.	Worked effectively as a team in designing and implementing the system
9	PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences	Presented project through reports, PPT, and demonstrations effectively
10	PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	Managed project planning, cost-effective design, and implementation
11	PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).	Learned and applied new technologies like IoT and carbon capture techniques

		Program Specific Outcome (PSOs)	
	PSO1		Apply knowledge of electronics and communication engineering to design and develop IoT-based systems for real-time monitoring and control applications.
	PSO2		Utilize modern tools and technologies to develop innovative solutions addressing environmental and sustainability challenges.

7.2 SDG Mapping

The following Sustainable Development Goals (SDG) are mapped for the project titled “IoT-Enabled CO₂ Capture and Carbonation Curing System for Sustainable Precast Concrete Blocks”,

S.NO	SDG	SDG	Mapped with the Project
1	SDG1	No Poverty End poverty in all its forms everywhere	
2	SDG2	Zero Hunger End hunger, achieve food security and improved nutrition and promote sustainable agriculture	
3	SDG3	Good Health and well-being Ensure healthy lives and promote well-being for all at all ages	
4	SDG4	Quality Education Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	
5	SDG5	Gender Equality Achieve gender equality and empower all women and girls	
6	SDG6	Clean water and Sanitation Ensure availability and sustainable	

		management of water and sanitation for all	
7	SDG7	Affordable and Clean Energy Ensure access to affordable, reliable, sustainable and modern energy for all	
8	SDG8	Decent work and Economic Growth Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	
9	SDG9	Industry, Innovation and Infrastructure Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation	This project promotes sustainable industrial innovation by integrating IoT-based monitoring with CO ₂ capture and carbonation curing in precast concrete production using fly ash for efficient and low-cost infrastructure development.
10	SDG10	Reduced Inequalities Reduce inequality within and among countries	
11	SDG11	Sustainable Cities and Communities Make cities and human settlements inclusive, safe, resilient and sustainable	
12	SDG12	Responsible Consumption and production Ensure sustainable consumption and production patterns	
13	SDG13	Climate Action Take urgent action to combat climate change and its impacts	This project contributes to climate action by reducing CO ₂ emissions through carbon capture and utilization in concrete curing and by lowering cement usage with fly ash, thereby promoting carbon sequestration and sustainable construction.

14	SDG14	Life below water Conserve and sustainably use the oceans, seas and marine resources for sustainable development	
15	SDG15	Life on Land Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss	
16	SDG16	Peace, Justice and Strong Institutions Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	
17	SDG17	Partnerships for the goals Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development	

CONFERENCE CERTIFICATION

We have Completed the Second International Conference on Engineering, Science and Management (ICESM 2026) organized by the Guru Gobind Singh College of Engineering and Research Centre, Nashik, Maharashtra, India & Research Hub, Coimbatore, Tamil Nadu, India on 10.04.2026 & 11.04.2026



